

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:BE01000021****Date:02-01-2025****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed

		Marks
Q.1	(a) Define: (1) Stress (2) Strain and (3) Young's modulus.	03
	(b) The critical temperature for a metal with isotopic mass 199.5 is 4.185 K. Calculate the isotopic mass if the critical temperature falls to 4.133 K.	04
	(c) What is Cantilever? Obtain the expression for depression at free end of thin beam clamped horizontally at one end and loaded at other end.	07
Q.2	(a) Define:(1) Reverberation (2) Nano-materials (3) Absorption co-efficient	03
	(b) The volume of a hall is 475 m^3 . The area of wall is 200 m^2 , area of floor and ceiling each is 100 m^2 . If the absorption co-efficient of wall, ceiling and floor are 0.025, 0.02 and 0.55, respectively. Calculate the reverberation time for the hall.	04
	(c) Describe production of ultrasonic waves by Piezo-electric method. Give its advantages and limitations.	07
	OR	
	(c) What is simple harmonic motion? Obtain equation for kinetic energy and potential energy of simple harmonic oscillator. Show that at any time the sum of kinetic energy and potential energy remains constant.	07
Q.3	(a) An ultrasonic source of 0.09 MHz sends down a Pulse towards the seabed which returns after 0.55 sec. The velocity of sound in water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse.	03
	(b) Differentiate: Spontaneous and Stimulated emission.	04
	(c) What are Einstein's coefficients? Give relation between them and discuss the result.	07
	OR	
Q.3	(a) Define:(1) Meta stable state (2) Population inversion (3) Pumping	03
	(b) Write a short note on Scanning Tunneling Microscope (STM).	04
	(c) What is full form of LASER? Write the characteristics and applications of LASER.	07
Q.4	(a) State any three differences between interference and diffraction.	03
	(b) Calculate the minimum number of lines in a grating which will just resolve the lines of wavelength $5890 \text{ \AA}$ and $5896 \text{ \AA}$ in the second order.	04
	(c) Derive the time independent Schrödinger's wave equation.	07
	OR	
Q.4	(a) An unknown weight is attached to the lower end of wire of length 4 m, radius 0.7 mm, extends it by 0.8 mm. If $Y = 2 \times 10^{11} \text{ N/m}^2$. Find the unknown weight.	03
	(b) Write a short note on black body radiation.	04
	(c) Explain Heisenberg's uncertainty principle and derive it.	07
Q.5	(a) Define: (1) Critical Temperature (2) Critical Magnetic field (3) Critical Current density.	03
	(b) Write the applications of nano materials.	04
	(c) Describe the principle, construction and working of Michelson interferometer.	07

OR

- Q.5** (a) State and explain principle of superposition of waves. **03**
(b) What is interference? Differentiate Constructive and destructive interference. **04**
(c) Explain working of the p-n junction diode and state its applications. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:BE01000021****Date:03-06-2025****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Transition temperature of mercury of mass number 202 is 4.20 K. Find the transition temperature of isotope of mercury of mass number 204.	03
	(b) Define: (i) Interference of light, (ii) Diffraction of light, (iii) Fresnel diffraction. Also state Huygens' principle.	04
	(c)(i) Define nanomaterials and explain ball milling technique of synthesis of nanomaterials with suitable diagram.	03
	(c)(ii) Write any four differences between n-type and p-type semiconductors.	04
Q.2	(a) What is de-Broglie wave? Write the equation of de-Broglie wavelength and mention the names of each quantity involved in the equation.	03
	(b) Find the force required to increase the length of a wire of 10^{-6} m^2 area of cross-section by 50 % whose Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.	04
	(c) Explain torsional pendulum in brief with suitable diagram and derive the equation of twisting couple per unit twist of a wire.	07
	OR	
	(c) Explain: (i) Elastic hysteresis, (ii) Elastic after-effect, and (iii) Elastic fatigue.	07
Q.3	(a) Define: (i) Transverse waves, (ii) Ultrasonic waves, and (iii) Damped oscillations.	03
	(b) The volume of a room is 1500 m^3 . The wall area of the room is 250 m^2 , the floor and the ceiling area is 150 m^2 each. The average sound absorption coefficient for the wall, floor and the ceiling are 0.25, 0.06 and 0.80, respectively. Calculate the average absorption coefficient and the reverberation time.	04
	(c) Derive the differential equation of simple harmonic motion, also write its general solution and obtain the equation of velocity and acceleration of simple harmonic oscillator.	07
	OR	
Q.3	(a) Define: (i) Reverberation, (ii) Echo, and (iii) Echelon effect.	03

	(b)	A particle of mass 0.50 kg executes a simple harmonic motion. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of the motion. (force constant $k = 50 \text{ N/m}$)	04
	(c)	Explain the construction and working of piezoelectric method of ultrasonic wave production with suitable diagram.	07
Q.4	(a)	In a Young's double slit experiment, the two slits which are 0.5 mm apart are illuminated by the yellow light of wavelength 589.0 nm. Calculate the distance between two consecutive bright bands in the interference pattern observed on a screen 2 m away from the slits.	03
	(b)	Write Schrodinger's time independent wave equation and explain its each term. Also write Heisenberg's uncertainty principle in terms of position and momentum.	04
	(c)	Explain the formation of Newton's rings with suitable diagram and derive the equation of radius of bright and dark rings produced by the interference of reflected light.	07
OR			
Q.4	(a)	Calculate de-Broglie wavelength associated with (i) A man of 50 kg mass walking with the speed of 2 m/s, and (ii) An electron moving with the speed of $3 \times 10^5 \text{ m/s}$. Based on your answers obtained in above two cases, explain why the wave nature of the man can't be observed. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$, electron mass = $9.11 \times 10^{-31} \text{ kg}$)	03
	(b)	Define constructive and destructive interference with their condition in terms of path difference.	04
	(c)	Write a short-note on scanning tunneling microscope.	07
Q.5	(a)	In He-Ne gas laser, Ne atoms emit laser photons of wavelength 632.8 nm while electrons make transitions from E_3 state to E_2 state. Find the energy difference between these two states. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$)	03
	(b)	Write the properties of laser and explain in brief.	04
	(c)	Write the full form of laser and explain (i) Absorption, (ii) Spontaneous emission, and (iii) Stimulated emission with suitable diagrams.	07
OR			
Q.5	(a)	A ruby laser emits light of wavelength 694.3 nm. Calculate the frequency and the energy of the laser photons emitted. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$)	03
	(b)	What is population inversion (draw suitable diagram)? Explain why it is necessary for the production of laser?	04
	(c)	Obtain the relation between Einstein's coefficients.	07

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - I/II EXAMINATION - WINTER 2025

Subject Code: BE01000021/BE01R00021

Date: 20-12-2025

Subject Name: Physics

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Enlist any six applications of Laser.	03
(b) Summarize any four factors affecting the acoustics of a building and discuss their remedies.	04
(c) Derive relation between elevation in rectangular beam, its length, breadth, thickness and Young's modulus in case of uniform bending.	07
Q.2 (a) A screen is separated from double slit source by 2 m. The distance between the two slits is 0.040 mm. The 3rd order bright fringe on the screen is measured to be 6.0 cm from the central line. Determine the wavelength of light.	03
(b) Explain elastic hysteresis and elastic fatigue in brief.	04
(c) Explain various applications of ultrasonic waves in industry including NDT in detail.	07
OR	
(c) What is simple harmonic motion? Obtain equation for kinetic energy and potential energy of simple harmonic oscillator. Show that at any time the sum of kinetic energy and potential energy remains constant.	07
Q.3 (a) Find the lowest energy of an electron confined to move in a cubical box of length 0.14 nm.	03
(b) Discuss wave particle duality using de Broglie hypothesis and derive the relation between wavelength and velocity of microscopic particle.	04
(c) Explain Newton's rings experiment and derive equation for the bright and dark fringes.	07
OR	
(a) Calculate the resultant sound level when a 75 dB is added to 85 dB sound?	03
(b) Introduce Michelson interferometer and its working briefly.	04
(c) Based on dimensions, enlist the types of nanomaterial and discuss the PVD method with schematic diagram for synthesis of thin film.	07
Q.4 (a) A superconductor with isotopic mass of 198 has critical temperature 198 K. What will be critical temperature of second isotope with mass 202.	03

- (b) Apprise using necessary energy level diagram that why 4 level laser and not 3 level laser is required for continuous operation. 04
- (c) Derive the equation for energy levels of a particle enclosed in one dimensional potential box. 07

OR

- (a) Calculate the twisting couple on a solid shaft of 1.5 m length and 0.12 m diameter when it is twisted through an angle of 0.0104 rad. The coefficient of rigidity of the material is $9.3 \times 10^{10} \text{ N/m}^2$. 03
- (b) What is black body radiation? Deduce Wien's law and Rayleigh-Jeans law from Planck's black body radiation law. 04
- (c) Define following terms: (1) Active medium (2) Pumping (3) optical resonator (4) population inversion (5) metastable state (6) stimulated emission and (7) coherence. 07

- Q.5**
- (a) Using necessary circuit diagram analyze the working of Zener diode as a voltage regulator. 03
 - (b) Describe critical magnetic field and Meissner effect for superconductors with suitable diagrams. 04
 - (c) Explain construction and working of Ruby laser using schematic diagram and energy level diagram. 07

OR

- (a) Discuss Huygen's principle in brief. 03
- (b) What are extrinsic and intrinsic semiconductors? How are p-type and n-type semiconductors made? 04
- (c) Derive the formula for the intensity of light in Young's double slit experiment and discuss the case for maximum and minimum intensity. 07
